

Are deep features able to learn melanoma depth gradation when trained only for binary tasks?

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ABSTRACT

This study aims to analyze dermoscopic images to estimate the Breslow depth of melanoma, which is crucial for diagnosing and treating skin cancer. The study has two objectives: to improve current methods of estimating melanoma depth and to investigate whether dermoscopic features evolve as melanoma thickness increases.

To achieve these goals, a Convolutional Neural Network (CNN), with ConvNext as backbone, was designed and trained to predict melanoma thickness. Then, the relationships between the CNN's outputs and the continuous depth values from biopsies were explored to investigate possible gradations in deep features related to melanoma depth. First, statistical analyses were performed in the deep feature spaces. Secondly, deep features were transformed into Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) spaces.

Two main results can be reported. First, the model outperformed existing methods in the literature for classifying melanomas as superficial or deep. It achieved the highest recall (0.79) and F1 score (0.67), a good balance between precision and recall. As for the relationships between deep features and biopsy depth, a moderate correlation was found between melanoma depth and prediction probabilities, which was stronger in deeper melanomas. Both PCA and UMAP analyses revealed that intermediate-depth melanomas have deep features located in intermediate positions in the transformed feature space.

These findings highlight the capacity of deep features to distinguish between different depths of melanoma. The study provides new insights into the data distribution and suggests a potential methodology for enhancing the understanding of melanoma classification through deep learning models.

1. Introduction

Melanoma is the most dangerous type of skin cancer [1,2] due to its tendency to spread. Melanoma can spread quickly to other parts of the body and it is difficult to treat once it metastasizes. It is potentially deadly if not detected early. Melanomas can develop anywhere on the body and often resemble moles; they are usually black or brown, but can also be skin-colored, red, or even blue. Unlike Basal Cell Carcinoma (BCC), melanomas can rapidly change in size, shape, or color [3,4].

There are different ways of assessing melanoma's depth, such as the Clark Index [5] or Breslow Index [6]. In this work, the latter is used, being a crucial parameter in determining the stage of the cancer and guiding treatment decisions. The Breslow depth is measured in millimeters from the top of the granular layer (or from the base of the ulcer, if the cancer is ulcerated) to the deepest point of tumor invasion.

It is a strong predictor of prognosis: thicker tumors are associated with a higher risk of metastasis and poorer outcomes [7]. Argenziano et al. [8] proposed an index based on Breslow depth, which groups melanoma thickness into two groups: thin melanoma (< 0.76 mm) and thick melanoma (> 0.76 mm), when diagnosing with dermoscopic images.

Given the clinical importance of accurate determination of the depth of melanoma, as a critical parameter in cancer staging and treatment planning, this research aims to develop a sophisticated deep learning model that can predict melanoma depth using dermoscopic images.

Few works in the literature have focused on the automatic estimation of the melanoma thickness. Optical Coherence Tomography [9] is a promising technique for depth estimation. In this sense, Carvalho et al. explore the relationship between the vascular morphology of melanoma

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and its Breslow index [10] with dynamic OCT. The research found that melanomas with higher Breslow thickness often exhibit specific vascular patterns, such as dotted and serpiginous branching vessels. These findings suggest a link between vascular characteristics and tumor aggressiveness, enhancing our understanding of the biological behavior of melanoma and its potential for metastasis.

Neittaanmäki et al. used hyperspectral imaging to detect lentigo melanoma invasion [11]. A total of 32 lesions, with histologically confirmed melanoma, participated in the study. The main concern of the study is its small sample size, and the difficulty in the interpretation of the hyperspectral image by the practitioners.

Imaging technique such as photoacoustic imaging has been studied and implemented to aid in the detection and diagnosis of cutaneous melanoma. Specifically, the measured thickness of the melanoma matches well with histopathological result (0.99 correlation) [12].

High-Frequency Ultrasound (HFUS) allows the pre- and post-operative assessment of melanoma with a 3D image. Initially HFUS was reported to overestimate tumor thickness when compared to histological section [13,14]. However, recent studies have obtained greater accuracy [15,16].

These techniques have the disadvantage of not usually being available in daily clinical practice. In addition, the use and handling of the acquisition devices requires specific training by healthcare personnel. However, dermatoscopy is present in most dermatology consultations. In this regards, Polesie et al. [17] focused on evaluating the precision of international readers, including dermatologists and general practitioners, and two machine learning convolutional neural networks (CNN), in measuring the thickness of melanoma from 1056 dermoscopic images. An open Web-based diagnostic study revealed moderate overall accuracy in melanoma thickness assessment, where the collective response of human readers exceeded one of the CNNs but not a pretrained CNN. The findings emphasize the difficulties in accurately assessing the thickness of melanoma where they lack significant recall to detect the important cases of thick melanoma. It is also worth mentioning another work by the same first author [18], where a study on the characteristics of melanoma is performed, finding differences and similarities between in situ and invasive melanoma.

Another contribution to the modeling and prediction of the thickness of melanoma is given by Saéz et al. [19], where instead of Deep Learning-based approaches, machine learning techniques are used, by relying the importance of classification on a set of hand-crafted features. These features are grouped on the basis of color, shape and texture. The study only uses 250 dermoscopic images of a moderately unbalanced dataset from Edra Atlas [20], the model achieves a 0.78 accuracy and a recall of 0.6.

Furthermore, Hernández-Rodríguez et al. [21] emphasize the efficacy of deep transfer learning (DTL) in differentiating between in situ and invasive melanomas according to the Breslow thickness. This finding highlights the potential of DTL as a robust adjunct tool for dermatologists. With a dataset of 1315 images, this work achieves a 0.75 accuracy and 0.58 F1 score with one of their models while others get up to 0.65 recall. The main drawbacks of this work are their highly unbalanced results, where the models are biased towards the majority class. They found that DTL approaches can even outperform dermatologists in Breslow thickness classification.

More recent works, such as Szijártó et al. [22] delve into the multiclass classification but do not achieve competitive results, and their datasets, of less than 250 images, are not up to the task.

As shown above, current models in the literature often fail to generalize well across diverse datasets and lack robust interpretability. This research seeks to overcome these limitations through targeted efforts in two main directions. First, we develop a more robust and powerful methodology for training convolutional neural networks. This optimization is based on several techniques that have been proposed in other applications, but have not been applied to this problem before. Secondly, in an attempt to contribute to interpretability, an insight into

Table 1

Class distribution across datasets, showing the significant imbalance in the ISIC dataset compared to the more balanced Edra Atlas and private datasets. The combined dataset remains skewed towards the thin class, highlighting the need to address class imbalance for effective model training.

ISIC		Edra Atlas			
	Samples	%		Samples	%
Thin	385	91.67	Thin	177	66.54
Thick	35	8.33	Thick	89	33.46
Total	420		Total	266	
Private 1		Private 2			
	Samples	%		Samples	%
Thin	168	58.54	Thin	107	56.61
Thick	119	41.46	Thick	82	43.39
Total	287		Total	189	
Full data set					
	Samples				
			%		
Thin	837		72.05		
Thick	325		27.95		
Total	1162				

the relationships between deep features and the continuous value of melanoma thickness is proposed, even though the deep neural network is only trained for binary classification.

Section 2 is devoted to the description of the methodology. Section 3 contains all the necessary details to reproduce the proposed system and the results obtained. In addition, an ablation study is also performed to analyze how the performance is improved when each of the proposed strategies are incorporated to the system. Section 3.2 discusses the results. Finally, some brief conclusions are presented in Section 4.4.

2. Methodology

For clarity, this section is divided into three distinct subsections. First, a description of the database used in this study and its characteristics is presented. Second, details about the deep learning techniques employed and the main training strategies used that most critically affected the melanoma thickness estimation performance. In the third subsection, new methods for clinical interpretability designed for this study are described.

2.1. Dataset

The full dataset consists of a combination of different sources. The first subset comes from the ISIC data collection [23]. It consists of images with thickness information, from which a total of 420 images were obtained. A detailed description of the legal terms of sharing data is described in <https://www.isic-archive.com/privacy-policy>. The second subset consists of 266 from the Edra Atlas of Dermatology. Edra Atlas database was published in 2000 in CD format that had to be purchased. These images were made publicly available in 2018 [20]. The other subsets (Private 1 and 2) are provided by Hospital Universitario Virgen del Rocío, Seville, Spain, with a total number of 476 images. This study was performed in line with the principles of the Declaration of Helsinki. The Hospital Universitario Virgen del Rocío Review Board granted ethics approval. Informed consent was obtained from all subjects involved in the study. These images are available under request [21].

Table 1 summarizes the number of samples provenient from each database together with their classification as thin or thick according to a 0.76 mm threshold [8]. The table highlights notable differences in class balance across the datasets. The ISIC dataset is highly imbalanced, with 91.67% of samples in the thin category and only 8.33% in the thick category. In contrast, the Edra Atlas and private datasets are more

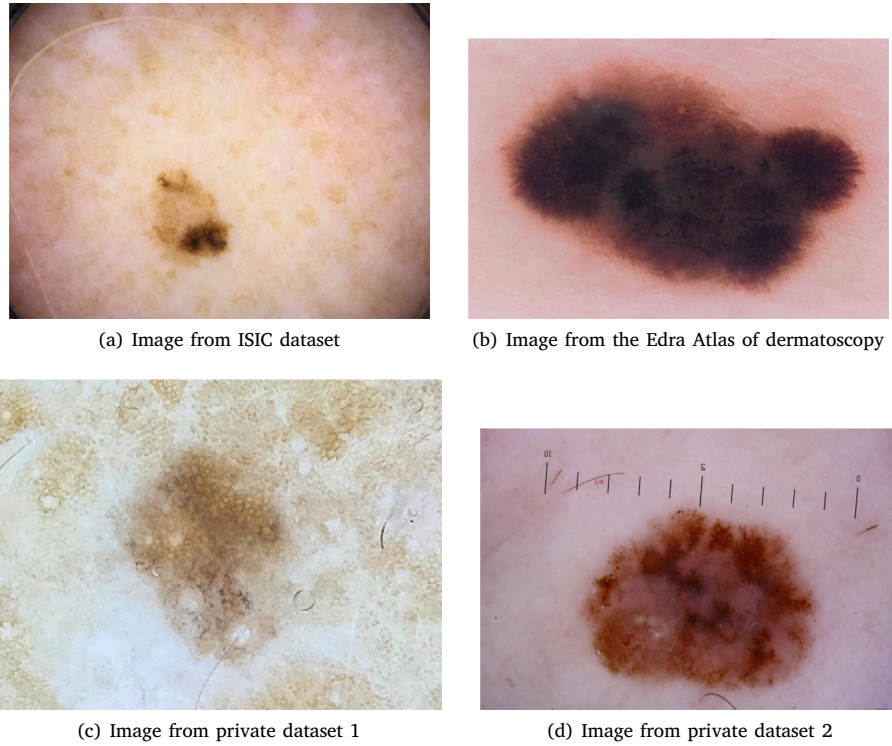


Fig. 1. Examples of dermoscopic images from different datasets, highlighting variations in acquisition techniques. (a) ISIC dataset image with standard lighting and clear focus. (b) Edra Atlas image with enhanced contrast and color saturation. (c) Private Dataset 1 image with lower resolution and diffuse lighting. (d) Private Dataset 2 image including measurement scales for clinical context. These differences reflect diverse imaging protocols and tools across sources.

balanced, with the thick class representing roughly one-third to nearly half of the samples.

The combined dataset contains 1162 samples, with 72.05% thin and 27.95% thick. While the imbalance persists, the inclusion of more balanced subsets mitigates its severity compared to the ISIC dataset alone.

Fig. 1 shows an example of each subset. Several differences between these images can be observed, such as image resolution, sharpness, or color. These different image characteristics are originated by varying acquisition conditions, e.g. where it was taken, the profile of the specialist who took the image or acquisition instrument.

2.2. Melanoma thickness prediction

In order to automatically predict the melanoma thickness, deep learning-based methods have been implemented.

In these models, dataset imbalance is important to consider, as it can bias model training towards the majority class, underscoring the need for techniques like resampling or class-weighted loss functions to ensure fair and accurate predictions.

A common problem in the field of deep learning or machine learning is that models tend to focus too much on the given data distribution, which does not represent the full diversity that exists in the real world. In this work, because the images come from different sources, the variability among samples is high, making the database more realistic and resilient. The models trained with data from such diverse sources allow for more robust classification with respect to future implementation with real medical systems [24,25].

An in-depth study of initializations, architectures, setting of parameter values, etc. have been addressed in order to outperform the state-of-the-art works. In particular, pre-trained backbones with data from the dataset ImageNet [26] were used. The pre-trained backbones, or also called feature extractors, were followed by a dense layer to

act as classifiers. These models were based of several different convolutional architectures, finding RegNet [27] and ConvNext [28] as the most promising ones.

The preprocessing step consists in resizing, normalizing, and cropping the samples, and on-line Data Augmentation (DA) is used before feeding the samples to the network as a way to improve generalization.

The training was done following a k-fold cross-validation scheme, supporting the training with sufficient data and a complete analysis of the entire dataset using it as a test. Training is split into a classifier-only tuning phase, followed by the fine-tuning of some of the last layers of the backbone alongside the classifier itself. It is widely used, as in [17,21], that the training of the pre-trained layers to fine-tune them is performed together with the classifier training, all at once. A more stable and accurate method is to perform the classifier training while freezing the entire backbone. When the classifier has reached convergence, fine-tuning is performed by unfreezing the last convolutional layers or modules of the backbone, with a lower learning rate to account for the increased number of parameters. This approach is especially important when working with small datasets.

The Focal Loss [29] was chosen as the loss function (see Eq. (1)). This function leverages the training by applying weighted loss to samples that are incorrectly classified. This technique has many advantages such as robustness against imbalanced datasets or ease in its hyper-parameter tuning. It is also compatible with class weightening and label smoothing, which are also used. Furthermore, cross-entropy loss can be derived from focal loss when setting the γ parameter to zero (Eq. (1)), which weights the importance of a misclassification in the loss. In the equation, the α represents the class weighting weights.

$$\begin{aligned} \text{FocalLoss}(p_i) &= -\alpha_i(1 - p_i)^\gamma \log(p_i), \quad \gamma = 0 \\ \text{CrossEntropyLoss}(p_i) &= -\alpha_i \log(p_i) \end{aligned} \quad (1)$$

The impact of some of the techniques is analyzed in Section 3.1, where an ablation study is performed. We identify three main additions

which have not been employed in the current literature for melanoma thickness prediction, which are the use of Focal Loss as a criterion, the application of the schedule-free optimizer, and the training of the classifier prior to the fine-tuning of the last convolutional kernels.

2.3. Deep feature analysis: XAI

Understanding and interpreting the results of neural networks is critical in healthcare. Explainable AI (XAI) builds trust by making AI decisions transparent and helping to debug and improve models. Most machine learning (ML) techniques, which are typically based of hand-crafted features related with clinical hints, are intrinsically interpretable models. But deep learning (DL) models need additional techniques to gather explanations to justify their decisions [30]. Techniques such as feature importance and saliency maps increase the explainability in AI systems, ensuring regulatory compliance, ultimately leading to the responsible and effective use of neural networks. However, in this work, the explanatory power is based on the analysis of deep feature distributions rather than activation maps. In this way, we also gain insight into how CNNs learn melanoma depth even without explicit thickness labels.

The deep features that are analyzed to interpret the results of the classifiers as well as to study possible gradation are the outputs of the last convolutional block. This output is formed by 768 maps sized 7×7 , but Global Average Pooling is applied to each map. Thus, the deep feature vector has a dimension of 768.

The methodology consists of an in-depth analysis of the data characteristics to support the decisions of the network, understanding its errors, trying to discover a certain degree of gradation in the samples, even if not explicitly trained for it. The two techniques used in this work are:

- **Linear regression analysis:** In this experiment, the outputs of the entire network, trained to distinguish between thin and thick melanomas, are compared to the actual thickness values of the lesions. The objective is to find if there is correlation between the output probabilities of the neural network and the actual depth of the melanoma (Breslow index). The hypothesis is that there should be some relationship between the predictions and the samples, where as the actual thickness of the sample increases, the predicted output for the class of thicker melanoma is higher. In this case, the model would learn this relationship indirectly, without the actual label for the actual thickness. This would also mean that the features characterizing one or the other class are progressively transformed as their thickness takes on more extreme values.
- **Deep features gradation:** Since these features are assumed to be directly related to the depth of the lesion, they can be compared to the bioscopy length to confirm the hypothesis. This technique is applied in two ways. First, Principal Component Analysis (PCA) is used to transform the set of features into a transformed space whose axes are ordered by their variances, thus capturing the information in fewer dimensions than using the deep features directly. Uniform Manifold Approximation and Projection (UMAP) [31] is a dimensionality reduction technique to highlight patterns in complex data, that preserves the global structure and local relationships of high-dimensional data. . With this representation, a hypothetical gradation would be detected.

These experiments on explainability are intended to find insightful information to support the clinical decision-making process.

2.4. Implementation details

The images from the datasets are processed in various ways before being fed into the model. First of all they are normalized to the range $\{0, 1\}$. Online data augmentation (DA) is also used in the training phase of the model, providing better generalization. The DA transformations selected for the training of the models are rotations, zooms, flips, and addition of Gaussian blur. No DA technique with respect to color is used to ensure robustness. All images are resized beforehand to 256×256 pixels.

As mentioned before, the underlying architectures for the models have been pre-trained with the ImageNet dataset [26], and are used following a deep transfer learning approach. Although several architectures have been tested in this work, the most successful one have been RegNet [27] and ConvNext Tiny [25].

To assess the performance of the model, a stratified k-fold cross-validation approach is employed. The selected k is 10, resulting in a fair balance between the time spent training and the quality of the validation.

Two main regularization techniques are used to prevent overfitting of the models. Firstly, dropout, which is a staple in neural network regularization which has been applied with a ratio of 0.3. The focal loss also acts as a regularization method by weighting samples by their classification complexity given the model as seen in (1), with γ set to 0.3.

Class weighting addresses the imbalance further by assigning more weight to underrepresented classes, improving the model's performance on these classes and preventing the scenario where the model only predicts the majority class to improve the loss. Even though the usual approach consists of setting these weights to the inverse of the frequency, this problem showed that those values did not grant enough relevance to the thicker class. Motivated by this, the inverse frequency value ($\times 2.6$ relative to the thinner class) was changed to $\times 5$, thus favoring the recall.

The Adam optimizer is chosen for its efficiency in handling sparse gradients and its adaptability to large datasets and high-dimensional spaces, used in conjunction with the scheduler-free scheme proposed in [32].

The training of the model is performed in stages, starting with the head and gradually fine-tuning (FT) deeper layers, allowing for a more controlled and effective learning process. Initially, the model adapts to high-level features using pre-trained weights, and then it refines its understanding based on the specific task. The learning rate for the first phase of the training was 10^{-3} , for 50 epochs, while the fine-tuning lasted the same number of epochs with a learning rate of 10^{-5} . Four convolutional blocks were fine-tuned in the case of both ConvNext and RegNet.

The batch normalization layers are not retrained, which helps stabilize the learning process and speed up the training. This approach leverages the normalization parameters learned from the pre-trained network, avoiding unnecessary adjustments during fine-tuning.

Regarding the models, several pre-trained weights shared by PyTorch for some architectures have been tested, highlighting the Regnet and ConvNext families, performing better than the others hosted by them (such as VGGs, MobileNets or Inceptions).

Lastly, the classifier used with these pre-trained backbones is just a dense layer, preceded by a Global Average Pooling layer after the last convolutional layer, with two outputs, one for each class.

For reproducibility, the code will be made available to readers upon request.

Table 2

Performance comparison of various classification models in related studies based on Recall, Precision, Accuracy, and F1-score. The proposed ConvNext model demonstrates the highest balanced performance, excelling in recall and achieving the best F1-score (0.67).

Model	Recall	Precision	Accuracy	F1
Sáez et al (LIPU) [19]	0.68	0.60	0.78	0.64
Polesie et al. (ResNet-50) [17]	0.44	0.71	0.84	0.54
Hernández et al. (ResNetV2) [21]	0.60	0.57	0.70	0.57
Hernández et al. (InceptionV3) [21]	0.65	0.53	0.65	0.54
Hernández et al. (EfficientNetB6) [21]	0.58	0.61	0.75	0.58
Nogales et al. (RegNet)	0.74	0.60	0.78	0.66
Nogales et al. (ConvNext)	0.79	0.60	0.79	0.67

3. Results

Table 2 summarizes the performance metrics of the proposed models and various classification models from different studies. In the comparison of the two models proposed in this work, which are based on the RegNet and ConvNext backbones, the ConvNext model achieved superior performance. Notably, the ConvNext model had the highest recall (0.79) and F1 score (0.67), demonstrating a strong balance between precision and recall. Overall, the results highlight the trade-offs between precision and recall among the models, with ConvNext emerging as the most balanced approach. The standard deviations of the performance metrics for this architecture are as follows: std_Recall = 0.07; std_Precision = 0.06; std_Accuracy = 0.03; std_F1 = 0.05. These values evidence the stability in performance of the proposed configuration.

Of the other methodologies in Table 2, Sáez et al. achieved balanced performance with their LIPU model, obtaining a recall of 0.68 and an accuracy of 0.78, although precision remained comparatively lower at 0.60. The dataset used in this work is also insufficient, featuring only a total of 250 samples. Polesie et al. using ResNet-50, achieved the highest accuracy (0.84) and precision (0.71) but showed a significant trade-off with a low recall of 0.44, indicating a tendency to miss many true positives. Hernández et al. explored multiple models (ResNetV2, InceptionV3, and EfficientNetB6) with varying results, generally performing moderately across all metrics.

Table 2 shows a slight improvement in the accuracy and F1 score of the proposed models compared to the others in the table, while the improvement in recall is more noticeable. However, caution should be exercised as the uncertainties shown by the standard deviation of the four performance metrics, in some cases, are comparable to the improvements. Specifically, the improvement of ConvNext over RegNet falls within ConvNext's standard deviation, and the improvement of ConvNext over [19] in F1 score is lower than ConvNext's standard deviation.

The good performance of model with ConvNext backbone may be due to the mix between the inherent performance of the CNN and its reduced number of parameters when using the tiny version. The number of deep features is smaller than that of other popular backbones, which may help the convergence of the dense classifier.

This improvement in performance can be also attributed to the careful application of various training techniques. Techniques such as k-fold cross-validation, online data augmentation, strategic class weighting, and the introduction of focal loss have played a pivotal role in optimizing the learning process. Furthermore, the adoption of advanced schedule-free optimizers like Adam and the use of dropout have contributed to a more robust and generalizable model. These methodological refinements have led to a notable enhancement in the model's predictive accuracy and overall reliability.

Table 3 shows the approximate computational costs of the different CNN backbones of the benchmark methods in Table 2 as estimated by torchvision.models subpackages [33]. As the table shows, the proposed

Table 3

Computational cost of CNN backbones employed by the methods listed in Table 2.

Backbone	Params (M)	FLOPs @224 (G)
ResNet-50 (Polesie et al. [17])	~25.6	~4.1
ResNetV2-50 (Hernández et al.[21])	~25.6	~4.1
InceptionV3 (Hernández et al.[21])	~23.8	~5.7
EfficientNet-B6 (Hernández et al.[21])	~43	~19
RegNetY-8GF (Nogales et al.)	~39	~8.0
ConvNeXt-Tiny (Nogales et al.)	~29	~4.5

Table 4

Performance metrics (Recall, Precision, Accuracy, and F1-score) for various ablation studies on ConvNext-based model configurations. Each row corresponds to a distinct experimental setup: baseline ConvNext, removal of focal loss, exclusion of the learning rate scheduler, and use of single-phase fine-tuning. The results highlight the trade-offs in model performance across different training adjustments.

	Recall	Precision	Accuracy	F1
Baseline	0.79	0.60	0.79	0.67
No focal loss	0.99	0.30	0.36	0.46
No scheduler	0.75	0.60	0.79	0.67
Single phase FT	0.76	0.56	0.76	0.64
No scheduler and single phase FT	0.63	0.67	0.79	0.65

method based on Tiny ConvNext requires the same number of parameters to be fixed and has a similar computational cost to the other benchmark methods. Thus, the important increase in recall is achieved without an increase in computational cost or training requirements. The method proposed by Sáez et al. [19] is not included in this comparison because its architecture is not based on a CNN backbone.

3.1. Ablation study

An ablation study was performed to analyze the effect of introducing the three techniques proposed in this paper to improve melanoma depth estimation, i.e. focal loss as the loss function, Adam schedule-free as the optimizer and two-phase training. For this purpose, each of them was removed separately. All these tests are performed with the ConvNext backbone.

The focal loss aims to apply a higher weight to samples that have been incorrectly classified, acting as a kind of reinforcement. To perform this test, the parameter γ of the focal loss is set to zero, which makes it equal to the cross-entropy loss, as previously shown in Eq. (1). The results can be seen in Table 4, where a significant drop in performance is observed. This may be due to the fact that without the focal loss the model tends to choose the most represented class.

When testing the optimizer, it can be seen that it does not affect the performance too much compared to the loss function, except for a slight drop in recall. This may be due to the small size of the dataset, related to the number of parameters updated in each iteration. However, it does help with convergence, making the descent to the optimum smoother and allowing higher learning rates.

Finally, instead of the two-phase, training consisting of training only the head and then fine-tuning the deeper layers of the pre-trained backbone, both parts are trained at the same time. This is useful for convergence speed, but it makes the models perform worse. In this case, the overall performance is slightly worse.

3.2. Correlation analysis of the network outputs

As explained in Section 2, the training of the CNNs was carried out considering two classes: thin (< 0.76 mm) and thick (> 0.76 mm). However, the depth of the biopsy is a continuous value, measured in millimeters. Therefore, a possible relationship between the continuous probability of the network output and this continuous depth was analyzed.

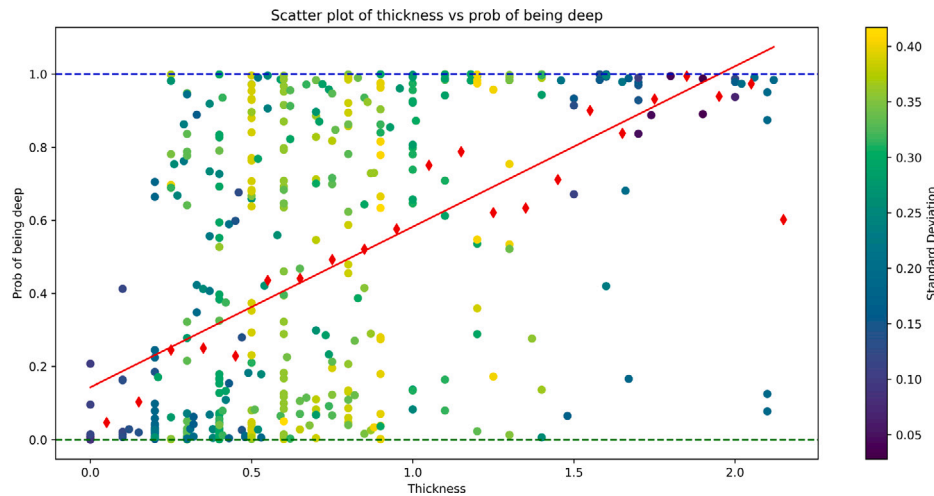


Fig. 2. Scatter plot and regression analysis of predicted probabilities for deepness versus real thickness, highlighting the influence of standard deviation on prediction consistency and class overlap in the intermediate thickness range.

Table 5

Segmented R^2 values indicating the strength of correlation between predicted probabilities and real thickness across different ranges, revealing weak model performance in the intermediate range (0.4 to 1 mm).

Ranges [mm]	R^2 Value
0 to 0.4	0.38
0.4 to 1	0.08
more than 1	0.44

Following this idea, linear, second-order and third-order regressors were trained to fit the biopsy depth values with the outputs of the neural network, i.e. the probability values of belonging to a specific class. The coefficients R^2 for these curves are low, a 0.25 value for linear regression, which does not improve for the higher-order ones, which stays at 0.26. The regression plot is shown in Fig. 2. The red line represents the linear regression line, where the red markers represent the mean of the points belonging to each 0.1 mm interval. The colors of the data samples represent different standard deviations as codified in the colorbar placed on the right of the figure. Data samples belonging to a thickness range between 0.4 mm and 1 mm are more scattered. This suggests that the samples belonging to this range could have mixed features from both deep and not deep thicknesses. To quantify this issue, the R^2 values for each thickness range are calculated and shown in Table 5, ensuring that there is a large difference between segments.

3.3. Analysis of transformed deep features

As proven in the previous subsection, melanomas with depths in the range 0.4 mm and 1 mm could present mixed features of thin and thick melanomas. This leads us to try to find a more suitable transformed space, where we can observe a greater separation between melanomas of different depths. To do this, deep features are projected into two subspaces in order to analyze how the distribution of samples would manifest itself.

The first subspace is obtained by selecting the most relevant PCA features. PCA provides a set of features ordered by their variance with respect to the samples. For visualization purposes, only the first two most relevant features are selected. Note that the term *relevant* in this case does not refer to the features with higher variance, but to those that make the interclass variance higher. In this case, these features are the first and the fourth ones.

The second approach was to project the set of deep features onto the axes given by a UMAP transformation. The reason for these choices

is to have two different bases on which to project the features, in the case of PCA, a general base that is not influenced by the labels of the models, only by the features. In the case of UMAP, to have a non linear transformation which attends to both global and local distances.

Figs. 3 and 4 show the distribution of the samples on the PCA and UMAP axes, respectively. The first and second subplots show the data projected onto the axes of the higher variance basis generated by PCA. In the case of UMAP only two axes are generated. In the first, the colors represent the original labels, with 0.76 mm set as the threshold between them. In the second, the samples belonging to the interval 0.4 mm and 1 mm are shown in gray, revealing the noisy characteristics of this group. Finally, the third plot shows ellipses fitted to the points to visualize the position of the distributions as well as their means. It can be clearly seen that the deep and not-deep classes are relatively separated, which is not the case for the intermediate range, as these samples seem to be more scattered. The mean of the distributions retains the slight gradation by thickness studied in the previous subsection, but the variance, which is easier to visualize in the ellipse plot, causes the distributions to overlap. It is important to note that in PCA, the gray ellipse covers a large part of the other two, supporting the hypothesis that samples in this interval have characteristics from the other two. The case of UMAP is very similar, but being a little bit less overlapped.

Finally, Fig. 5 shows the same distribution as in the PCA projection, but expressed only on the axis with the highest variance (most relevance). Gaussian distributions have been fitted to the sample distributions to show how they relate to the data, which is shown at the bottom with less opacity. The extreme groups, with the higher and lower thicknesses, fit the distribution closely, while the middle group does not fit as well.

4. Discussion

There are few papers in the literature that have addressed the problem of classifying melanomas as in situ or invasive or as superficial or deep. However, to the best of our knowledge, none of these works have analyzed whether the deep features extracted from the images in their binary classification task have a continuous evolution as the depth of the lesions evolves from superficial to deep melanoma. This analysis and the relationship found could be the beginning of answering other questions: 1. Explain the binary classification of the network from the deep features - a question already explored in the literature. 2. Improve the accuracy of the binary classification by exploiting the grading information found. 3. Identify possible correlations between deep features and image features to help dermatologists find a consensus of patterns from which to estimate a depth scale.

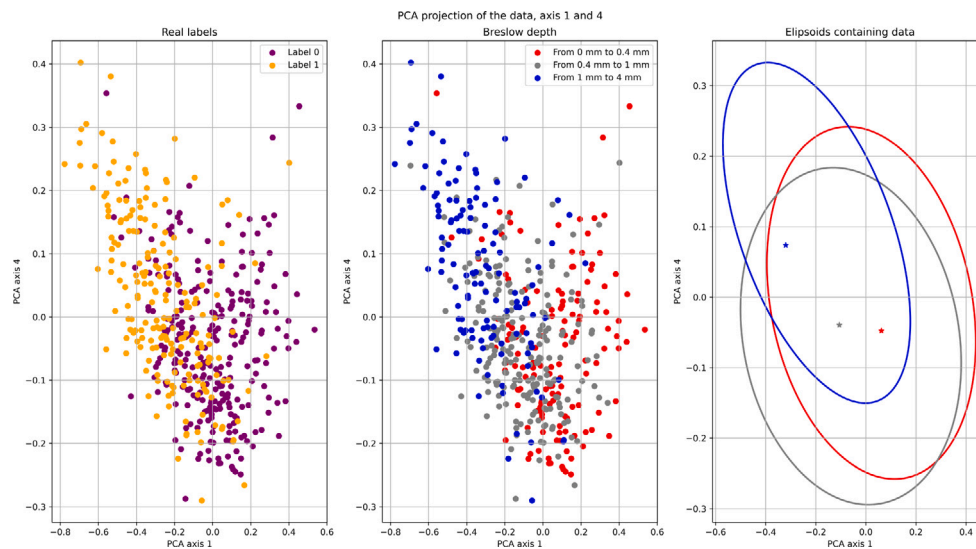


Fig. 3. PCA projection of deep features: Visualization of real labels (left), predicted probabilities with decision boundaries (center), and ellipsoidal clusters representing class separability (right), emphasizing the overlap between classes and the model's challenges in distinguishing intermediate features.

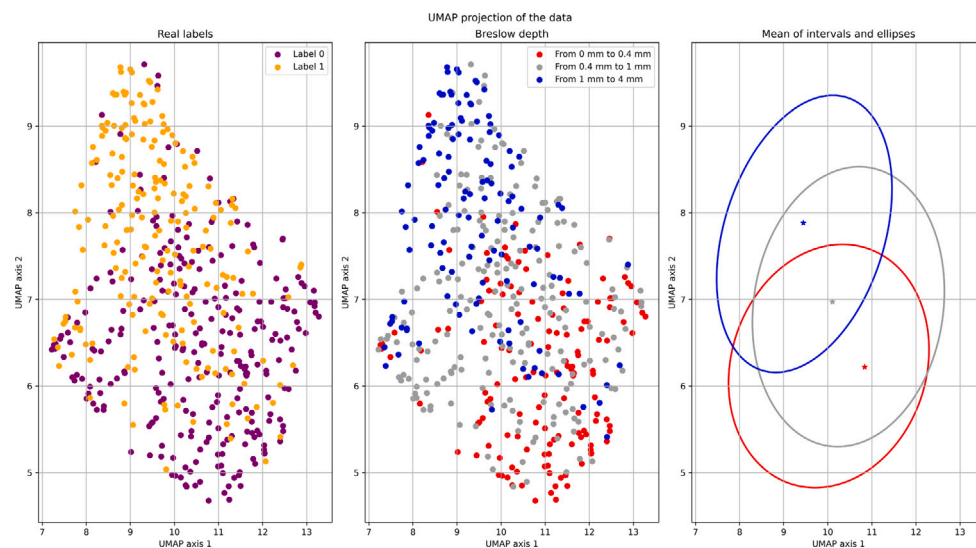


Fig. 4. UMAP projection of deep features: Visualization of real labels (left), predicted probabilities with decision boundaries (center), and ellipsoidal clusters representing class separability (right), emphasizing the overlap between classes and the model's challenges in distinguishing intermediate features.

4.1. Contributions and interpretation of results

The primary objective of this study was to assess whether convolutional neural networks (CNNs) designed for binary classification of melanoma thickness could simultaneously provide meaningful estimates of melanoma depth. By linking the binary classification task with depth estimation, the study explored whether models trained to distinguish between thick and thin melanomas inherently learn finer gradations in thickness.

The results, particularly for models based on ConvNext and RegNet architectures, revealed significant improvements over existing methods in estimating melanoma thickness. The analysis revealed a significant improvement in recall, with our ConvNext model achieving the highest balanced performance across all key metrics, including an F1 score of 0.67 and a recall of 0.79. This is particularly important in medical diagnostics, where prioritizing recall reduces the risk of false negatives, minimizing the likelihood of undetected thick melanomas with severe clinical implications. While models such as ResNet-50 achieved high precision, their low recall highlights a critical trade-off that could result

in many missed positive cases. ConvNext's superior performance can be attributed to the careful use of focal loss, class weighting, and fine-tuning techniques, which addressed class imbalance and improved generalization. Class imbalance is a common issue in medical datasets, and the combination of these methods enhanced the model's ability to detect minority class samples effectively. Additionally, transfer learning from pre-trained models allowed better feature extraction, despite the limited dataset size. In summary, the ConvNext model demonstrated a well-balanced performance across recall, precision, and F1 score, making it a robust approach for clinical applications where sensitivity is critical.

Correlation analysis showed moderate relationships between network outputs and biopsy melanoma thickness. Specifically, the linear regression model yielded a mean R^2 value of 0.25, reflecting a modest predictive ability of the network across the range of melanoma thicknesses. However, when the analysis was segmented into different thickness ranges, a more nuanced picture emerged. The R^2 value for the thin range (0 to 0.4 mm) was 0.38, indicating a relatively stronger correlation compared to other ranges. In contrast, the intermediate

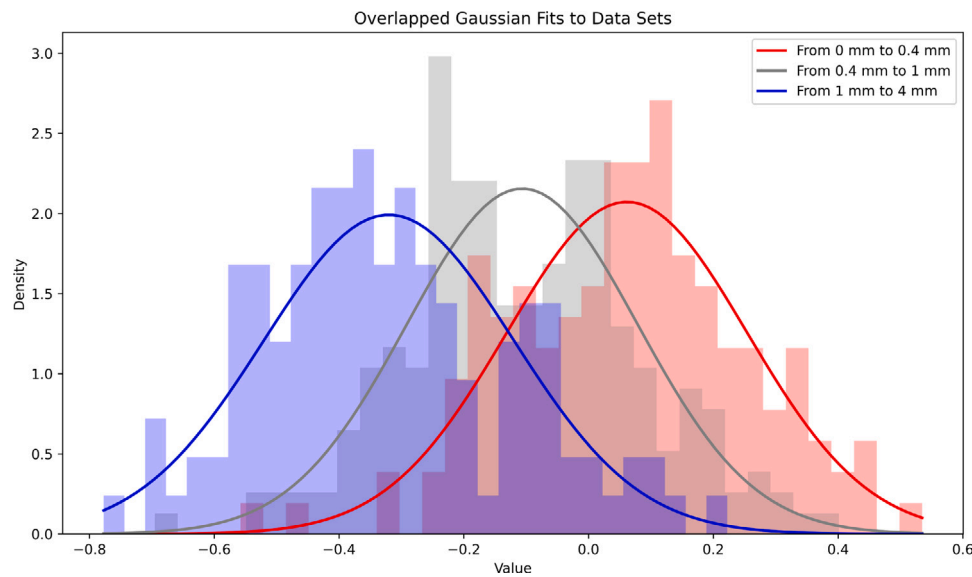


Fig. 5. Gaussian fits to one-dimensional PCA projections across thickness ranges, illustrating the density distribution and overlap between ranges, with notable ambiguity in the intermediate (0.4–1 mm) range.

range (0.4 mm to 1 mm) had a significantly low R^2 value of 0.08, highlighting the weak correlation and considerable overlap between features from the thin and thick categories. The thick range (over 1 mm) showed an R^2 value of 0.44, suggesting a better correlation in this segment, probably because thicker melanomas have more distinct features. This segmented analysis highlights the challenge posed by the intermediate range, where melanomas often have overlapping characteristics, leading to ambiguity in classification. This behavior is consistent with clinical observations, where melanomas of intermediate depth are known to display heterogeneous features that make clear categorization difficult [34]. These findings emphasize the need for specialized strategies to handle intermediate cases more effectively in future models.

A key contribution of this work is the application of explainable AI (XAI) techniques to explore the decision-making process of CNNs. Feature extraction, selection, and explanation from medical images form the basis of radiomics and the discovery of image biomarkers. Dimensionality reduction is a key step in this process to avoid the curse of dimensionality, which occurs as the number of features increases [35]. Thus, dimensionality reduction has been employed in this work to provide explainability. By employing dimensionality reduction methods such as PCA and UMAP, the study provided a visualization of deep features, revealing distinct clusters for thin and thick melanomas. However, the intermediate range showed significant overlap with both classes, supporting the hypothesis that CNNs trained on binary tasks can still learn underlying gradations in the data, albeit with reduced clarity in ambiguous cases. This gradation, although not explicitly trained, provides valuable insight into the behavior of deep networks in medical image analysis, suggesting that deep features may encode more information about melanoma thickness than just binary labels. These findings underscore the importance of interpretability in AI-driven medical applications, where understanding model decisions can increase trust and adoption in clinical settings.

4.2. Comparison with prior work

Compared to previous studies, such as those by Polesie et al. [17] and Hernández-Rodríguez et al. [21], the proposed approach demonstrated superior performance in both recall and F1-score. While Polesie et al. achieved the highest accuracy, their model suffered from a low recall, highlighting a trade-off that is less desirable in medical applications. In addition, the high accuracy reported by Polesie et al. can

be partly attributed to the high class imbalance in their dataset, which may have biased the model towards correctly predicting the majority class at the expense of recall. The inclusion of explanatory methods in this study further distinguishes it from previous work by providing insight into how the models learn and discriminate melanoma thickness, thereby improving interpretability and potential clinical adoption.

In addition, the study of the relationship between deep features of individual samples and their pattern is not sufficiently explored in the literature [36], although there are works that indirectly probe these relationships. In particular, all the works exploring the latent space of Variational Autoencoders [37] (VAE), or autoencoders in general, show some similarity to our approach, as they explain the network behavior in terms of features [38,39].

4.3. Limitations and future work

Despite the promising results, several limitations must be acknowledged. First, the dataset used, although diverse, remains relatively small compared to large medical imaging datasets. This limits the generalizability of the findings, especially when applied to new populations with different imaging conditions. Second, class imbalance, although addressed by techniques such as class weighting and focal loss, may still have affected the performance of the model. Further studies with larger and more balanced datasets are needed to validate these findings.

Additionally, the use of binary classification for a problem that inherently involves continuous gradation may limit the model's ability to fully capture the nuances of melanoma thickness. A key challenge in this area is the scarcity of high-quality datasets with detailed dermatological depth annotations, which limits the ability to develop more sophisticated models. Future work could explore regression-based approaches or multi-class classification to better represent the continuous nature of melanoma depth, but significant efforts will be required to collect and curate larger datasets that accurately reflect the full spectrum of melanoma thickness. For example, including data from different demographics groups or training jointly different modalities, with methodologies such as CLIP [40].

Finally, it is an important point to correlate melanoma depth with dermoscopic patterns. And this will be the immediate future work. In the present work, we have investigated how deep features behave in relation to melanoma depth, but we have not looked at the clinical significance of these features, i.e. their relationship with the observed dermoscopic features. However, our study has served as a starting point for this research.

4.4. Conclusions

This study makes two main contributions: First, it improves melanoma depth classification over state-of-the-art work. Second, it offers new insights into the possible gradation of melanoma dermoscopic features with depth.

The first contribution has an important clinical implication, as it supports physicians in their daily dermatology consultations. Estimating the depth of melanoma greatly influences surgical planning decisions.

This improvement over state-of-the-art melanoma depth classifiers is achieved thanks to a robust methodology involving pre-trained convolutional neural networks (CNNs) and rigorous validation methods, such as k-fold cross-validation. The inherent diversity of the database also contributes to this improvement.

Second, a novel study on melanomas in the intermediate depth range and its correlation with deep neural network outputs in both the original and transformed spaces (PCA and UMAP) has provided new insights into the gradation of dermoscopic melanoma features with depth. Specifically, although the deep features were trained for binary classification, they revealed intermediate clusters for intermediate biopsy depths in the transformed space. This finding is significant because it suggests a gradation in melanoma lesion visual features.

However, this study establishes an important direction for future research.

A multimodal approach and new multimodal AI models could improve classification results. Combining dermoscopic image information with clinical information and other image modalities, such as thermal images, could enhance the model's predictive capability.

Another important future direction is to explore the relationship between the dermoscopic features reported by dermatologists and the deep features. This will allow us to explore links between biopsies, dermoscopic features, and deep features. This will enable the development of physics-informed neural networks by embedding clinical criteria into the model loss function. Another important complementary analysis will be comparing the explanatory power of these deep learning (DL) features versus handcrafted features [35].

CRediT authorship contribution statement

Miguel Nogales: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Begoña Acha:** Writing – review & editing, Visualization, Validation, Supervision. **Fernando Alarcón-Soldevilla:** Formal analysis, Data curation, Conceptualization. **José-Juan Pereyra-Rodríguez:** Formal analysis, Data curation, Conceptualization. **Carmen Serrano:** Writing – review & editing, Visualization, Validation, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used DeepL Write in order to check the text grammar. After using this tool/service, the authors reviewed and edited the content as needed and hence take full responsibility for the content of the submitted article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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